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**Thapar Institute of Engineering & Technology, Patiala**
Department of Computer Science and Engineering
**Auxiliary Examination**

**BE (COE/CSE)** Course Code: **UCS510/ UCS319**
Semester: 5th & 3rd Course Name: **Computer Architecture and Organization**

Date: 19-02-2025 Max. Marks: 100
Time: 05:30 PM – 08:30 PM Name of Faculty: Dr. Anamika Sharma

***Attempt all questions. Please be precise and concise while writing. Assume any missing data.***

**Q1** Simplify the following Boolean function in sum-of-products form using a four-variable map. Draw the logic diagram with (a) AND-OR gates (b) NAND gates. **(10)**

$F(A, B, C, D) = \sum(0, 2, 8, 9, 10, 11, 14, 15)$

**Q2** Differentiate between IEEE-754 single-precision and double-precision representation formats for floating point numbers. Convert the following numbers in IEEE-754 floating point 32-bit short precision format. **(10)**

i. $(B21E)_{16}$
ii. $(-51.625)_{10}$

**Q3** Describe different pipelining hazards and how they can impact the performance of a pipelined processor. Explain with suitable examples of situations where these hazards may occur and how they can be resolved. **(10)**

**Q4** How does DMA (Direct Memory Access) bypass the CPU and speed up the memory operation? Explain your answer in detail considering its block diagram, registers, and mode of transfer. **(10)**

**Q5** Explain in detail the significance of Booth's Algorithm over the traditional algorithm of multiplication. Assume 5-bit registers that hold the signed numbers. Show the step-by-step multiplication process using the Booth Algorithm when the multiplicand $(-15)_{10}$ is multiplied with the multiplier $(13)_{10}$. **(10)**

**Q6** Show the content in hexadecimals of registers PC, AC, AR, DR, IR, and SC of the basic computer. Given the initial content of PC is F89, the content of AC is 7EC3. The content of memory at address F89 is 8BCF. The content of memory at address BCF is A189. The content of memory at address 189 is FFFF. Give the answer in the table with six columns. Repeat the problem six more times starting with an opcode of another memory reference instruction. **(10)**

**Q7** Explain in detail the different addressing modes for an instruction. How effective address is calculated for each addressing mode? **(10)**

**Q8** A system has a cache with 64 blocks and uses direct mapping. If the main memory has 4096 blocks, how many blocks can map to the same cache location? Derive the formula for calculating the number of bits required for tag, index, and offset in direct-mapped cache. **(10)**

**Q9** Consider a processor with memory unit of having 4096 words and each word of size 16 bit. Explain the special purpose registers namely PC, AR, DR, TR, IR, AC, INPR and OUTR along with their size for the given configuration in detail. **(10)**

**Q10** Design an arithmetic circuit with one selection variable S and two 4-bit data inputs A and B. The circuit generates the following four arithmetic operations in conjunction with the input carry Cin. **(10)**

| S | $C_{in}$=0 | $C_{in}$=1 |
|---|---|---|
| 0 | D=A+B | D=A+1 |
| 1 | D=A-1 | D=A-B |